Chapter 6

Adjunctions

6.1 Definition

Definition 6.1.1 (Adjunction)**.** [1] An *adjunction* $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle$ consists of two functors and two natural transformations as shown:

$$\mathcal{X} \underset{\boldsymbol{G}}{\overset{\boldsymbol{F}}{\rightleftarrows}} \mathcal{A} \qquad \eta : \boldsymbol{id}_{\mathcal{X}} \stackrel{\cdot}{\longrightarrow} \boldsymbol{GF} \qquad \varepsilon : \boldsymbol{FG} \stackrel{\cdot}{\longrightarrow} \boldsymbol{id}_{\mathcal{A}}$$

such that $\varepsilon_{\boldsymbol{F}\boldsymbol{X}} \circ \boldsymbol{F}\eta_X = id_{\boldsymbol{F}X}$ for every object X in $\mathcal{X}$, and $\boldsymbol{G}\varepsilon_A \circ \eta_{\boldsymbol{G}A} = id_{\boldsymbol{G}A}$ for every object A in $\mathcal{A}$.

$\boldsymbol{F}$ is called *left adjoint* to $\boldsymbol{G}$, and $\boldsymbol{G}$ is *right adjoint* to $\boldsymbol{F}$ (notation $\boldsymbol{F} \dashv \boldsymbol{G}$). η is called the *unit* of the adjunction and ε is called *counit*. The equations on the unit and counit are called the *triangular identities*.

Example 6.1.2. In a Cartesian closed category, $\_ \times A \dashv \_^A$ holds for every object A, with unit $\lambda(id_{\_ \times A})$ and counit $ev_{A,\_}$.

Exercise 6.1.3. Confirm that, in the adjunction above, unit and counit are natural transformations and also satisfy triangular identities.

Exercise 6.1.4. Given an adjunction as above, construe $\boldsymbol{F}$ as a functor $\boldsymbol{F}$:

[1] McLarty (1992, p.89)

$\mathcal{X}^{op} \longrightarrow \mathcal{A}^{op}$, and similarly for $\boldsymbol{G}$. Show that $\boldsymbol{F}$ is right adjoint to $\boldsymbol{G}$ with unit ε and counit η.

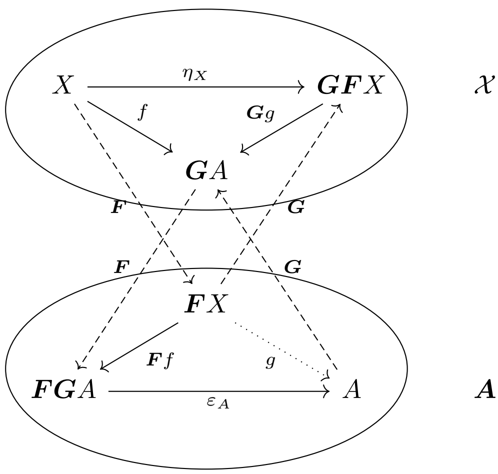
Remark 6.1.5. Thus any theorem on all left adjoints or all counits has a dual theorem on all right adjoints or all units.

6.2 Adjunction as a selection of universal arrows

Theorem 6.2.1. [2]Given an adjunction $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle$ as above, for every $\mathcal{X}$-object X, a pair of $\mathcal{A}$-object $\boldsymbol{F}X$ and $\mathcal{X}$-arrow $\eta_X : X \longrightarrow \boldsymbol{GF}X$ is universal from X to $\boldsymbol{G}$. Specifically, the adjunct of any $f : X \longrightarrow \boldsymbol{G}A$ with respect to η_X is $\varepsilon_A \circ \boldsymbol{F}f$.

Dually, for every $\mathcal{A}$-object A, a pair of $\mathcal{X}$-object $\boldsymbol{G}A$ and $\mathcal{A}$-arrow ε_A : $\boldsymbol{FG}A \longrightarrow A$ is universal from $\boldsymbol{F}$ to A, and the adjunct to any $g : \boldsymbol{F}X \longrightarrow A$ is $\boldsymbol{G}f \circ \eta_X$.

Remark 6.2.2. Namely, for any $\mathcal{X}$-arrow f, there is a unique $\mathcal{A}$-arrow g that makes the following diagram commute, and for any $\mathcal{A}$ arrow g, there is a unique $\mathcal{X}$-arrow f that makes the following diagram commute.



[2]McLarty (1992, p.90)

Proof. It suffices to show that, for any $\mathcal{X}$-object X, $\mathcal{A}$-object A and $\mathcal{X}$-arrow $f : X \longrightarrow \boldsymbol{G}A$, the following diagram commmutes:

$$\begin{tikzcd}
X \arrow[r, "\eta_X"] \arrow[dr, "f"'] & \boldsymbol{GF}X \arrow[d, "\boldsymbol{G}(\varepsilon_A \circ \boldsymbol{F}f)"] & \boldsymbol{F}X \arrow[d, "\varepsilon_A \circ \boldsymbol{F}f"] \\
& \boldsymbol{G}A & A
\end{tikzcd}$$

$$\begin{aligned}
\boldsymbol{G}(\varepsilon_A \circ \boldsymbol{F}f) \circ \eta_X &= \boldsymbol{G}\varepsilon_A \circ \boldsymbol{GF}f \circ \eta_X \\
&= \boldsymbol{G}\varepsilon_A \circ \eta_{\boldsymbol{G}A} \circ f && \text{(By naturality of } \eta\text{)} \\
&= id_{\boldsymbol{G}A} \circ f && \text{(By triangular identity)} \\
&= f
\end{aligned}$$

Uniqueness: Suppose that, for some $g : \boldsymbol{F}X \longrightarrow A$, $\boldsymbol{G}g \circ \eta_X = f$ holds. Then:

$$\begin{aligned}
\varepsilon_A \circ \boldsymbol{F}f &= \varepsilon_A \circ \boldsymbol{F}(\boldsymbol{G}g \circ \eta_X) && \text{(By premise)} \\
&= \varepsilon_A \circ \boldsymbol{FG}g \circ \boldsymbol{F}\eta_X \\
&= g \circ \varepsilon_{\boldsymbol{F}X} \circ \boldsymbol{F}\eta_X && \text{(By naturality of } \varepsilon\text{)} \\
&= g \circ id_{\boldsymbol{F}X} && \text{(By triangular identity)} \\
&= g
\end{aligned}$$

Thus $\varepsilon_A \circ \boldsymbol{F}f$ is the unique arrow that makes the triangle commute. □

Exercise 6.2.3. Draw the naturality square into the diagram above.

Theorem 6.2.4. If $\boldsymbol{F} \dashv \boldsymbol{G}$ and $\boldsymbol{F} \dashv \boldsymbol{G}'$, there is a natural isomorphism $i : \boldsymbol{G} \stackrel{\cdot}{\longrightarrow} \boldsymbol{G}'$. More specifically, if $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle$ is an adjunction and $\langle \boldsymbol{F}, \boldsymbol{G}', \eta', \varepsilon' \rangle$ is another, then there is a unique natural isomorphism i with $\varepsilon' = \varepsilon \circ \boldsymbol{F}i$. This is also the unique natural isomorphism satisfying the dual equation for the units.

The dual result holds for $\boldsymbol{F}$ and $\boldsymbol{F}'$ both left adjoint to $\boldsymbol{G}$.

Proof. By Theorem 5.1.3 and Theorem 6.2.1, for each $\mathcal{A}$-object A, there is a unique isomorphism $i_A : \boldsymbol{G}'A \longrightarrow \boldsymbol{G}A$ with $\varepsilon'_A = \varepsilon_A \circ \boldsymbol{F}i_A$. Thus we only need to show that these form the component of a natural transformation; that is, for every $h : A \longrightarrow B$ of $\mathcal{A}$, $\boldsymbol{G}h \circ i_A = i_B \circ \boldsymbol{G}'h$.

By Theorem 6.2.1, it suffices to show that the adjuncts of both sides coincides:

$$\begin{aligned}
\varepsilon_B \circ \boldsymbol{F}(\boldsymbol{G}h \circ i_A) &= \varepsilon_B \circ \boldsymbol{FG}h \circ \boldsymbol{F}i_A \\
&= h \circ \varepsilon_A \circ \boldsymbol{F}i_A && \text{(By naturality of } \varepsilon\text{)} \\
&= h \circ \varepsilon'_A && \text{(By definition of } i_A\text{)}
\end{aligned}$$

□

$$\begin{aligned}
\varepsilon_B \circ \boldsymbol{F}(i_B \circ \boldsymbol{G}'h) &= \varepsilon_B \circ \boldsymbol{F}i_B \circ \boldsymbol{FG}'h \\
&= \varepsilon'_B \circ \boldsymbol{FG}'h && \text{(By definition of } i_B\text{)} \\
&= h \circ \varepsilon'_A && \text{(By naturality of } \varepsilon'\text{)}
\end{aligned}$$

Lemma 6.2.5. Let a functor $\boldsymbol{F} : \mathcal{X} \longrightarrow \mathcal{A}$ have a selected universal arrow $\boldsymbol{G}A, \varepsilon_A$ to each object A of $\mathcal{A}$, when '$\boldsymbol{G}A$' names some object of $\mathcal{X}$ and shows that this object depends on A. Then there is just one way to extend $\boldsymbol{G}$ to a functor $\boldsymbol{G} : \mathcal{A} \longrightarrow \mathcal{X}$ so that the universal arrows become components of a natural transformation $\varepsilon : \boldsymbol{FG} \longrightarrow \boldsymbol{id}_A$.

Proof. For any $\mathcal{A}$-arrow $f : B \longrightarrow A$, let $\boldsymbol{G}f$ be the adjunct of $f \circ \varepsilon_B$, which is an arrow from $\boldsymbol{G}B$ to $\boldsymbol{G}A$.

$$\begin{tikzcd}
\boldsymbol{FG}A \arrow[rr, "\varepsilon_A"] & & A \\
\boldsymbol{FG}B \arrow[r, "\varepsilon_B"'] & B \arrow[ur, "f"'] &
\end{tikzcd}$$

Then we only need to show that $\boldsymbol{G}$ is a functor, and ε is a natural transformation from $\boldsymbol{FG}$ to $\boldsymbol{id}_A$. □

Exercise 6.2.6. Complete the above proof, following the proof of $\_^A$ being a functor.

Lemma 6.2.7. For every $\mathcal{X}$-object X, we define $\eta_X : X \longrightarrow \boldsymbol{GF}X$ to be the adjunct to $id_{\boldsymbol{F}X}$ relative to $\varepsilon_{\boldsymbol{F}X}$. Then arrows η_X form a natural transformation $\eta : \boldsymbol{id}_X \longrightarrow \boldsymbol{GF}$.

Proof. We will show that the following diagram commutes for every $\mathcal{X}$-arrow $f : X \longrightarrow Y$, namely, $\boldsymbol{GF}f \circ \eta_X = \eta_y \circ f$.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\eta_X"'] & Y \arrow[d, "\eta_Y"] \\ \boldsymbol{GF}X \arrow[r, "\boldsymbol{GF}f"'] & \boldsymbol{GF}Y \end{tikzcd}$$

By Theorem 6.2.1, it suffices to show that the adjuncts of both sides coincides:

$$\begin{array}{rcll} \varepsilon_{\boldsymbol{F}Y} \circ \boldsymbol{F}(\boldsymbol{GF}f \circ \eta_X) & = & \varepsilon_{\boldsymbol{F}Y} \circ \boldsymbol{FGF}f \circ \boldsymbol{F}\eta_X & \\ & = & \boldsymbol{F}f \circ \varepsilon_{\boldsymbol{F}X} \circ \boldsymbol{F}\eta_X & \text{(By naturality of } \varepsilon) \\ & = & \boldsymbol{F}f \circ id_{\boldsymbol{F}X} & \text{(By definition of } \eta_X) \\ & = & \boldsymbol{F}f & \end{array}$$

$$\begin{array}{rcll} \varepsilon_{\boldsymbol{F}Y} \circ \boldsymbol{F}(\eta_Y \circ f) & = & \varepsilon_{\boldsymbol{F}Y} \circ \boldsymbol{F}\eta_Y \circ \boldsymbol{F}f & \\ & = & id_{\boldsymbol{F}Y} \circ \boldsymbol{F}f & \text{(By definition of } \eta) \\ & = & \boldsymbol{F}f & \end{array}$$

□

Lemma 6.2.8. The natural transformation η and ε satisfy the triangular identities.

Proof. $\varepsilon_{\boldsymbol{FX}} \circ \boldsymbol{F}\eta_X = id_{\boldsymbol{F}X}$ follows from the definition of η.

To show $\boldsymbol{G}\varepsilon_A \circ \eta_{\boldsymbol{G}A} = id_{\boldsymbol{G}A}$, by Theorem 6.2.1 again, it suffices to show that the adjuncts of both sides coincides:

$$\begin{array}{rcll} \varepsilon_A \circ \boldsymbol{F}(\boldsymbol{G}\varepsilon_A \circ \eta_{\boldsymbol{G}A}) & = & \varepsilon_A \circ \boldsymbol{FG}\varepsilon_A \circ \boldsymbol{F}\eta_{\boldsymbol{G}A} & \\ & = & \varepsilon_A \circ \varepsilon_{\boldsymbol{FG}A} \circ \boldsymbol{F}\eta_{\boldsymbol{G}A} & \text{(By naturality of } \varepsilon) \\ & = & \varepsilon_A \circ id_{\boldsymbol{FG}A} & \text{(By definition of } \eta_{\boldsymbol{G}A}) \\ & = & \varepsilon_A & \end{array}$$

$$\begin{array}{rcl} \varepsilon_A \circ \boldsymbol{F}(id_{\boldsymbol{G}A}) & = & \varepsilon_A \circ id_{\boldsymbol{FG}A} \\ & = & \varepsilon_A \end{array}$$

□

Theorem 6.2.9. Given a functor $\boldsymbol{F} : \mathcal{X} \longrightarrow \mathcal{A}$ and a selected object $\boldsymbol{G}A$ of $\mathcal{X}$ and unversal arrow $\boldsymbol{G}A, \varepsilon_A : \boldsymbol{FG}A \longrightarrow A$ for each object A of $\mathcal{A}$, $\boldsymbol{G}$ extends to a functor $\boldsymbol{G} : \mathcal{A} \longrightarrow \mathcal{X}$ in exactly one way such that the arrows ε_A form a natural transformation from $\boldsymbol{FG}$ to $\boldsymbol{id}_{\mathcal{A}}$. This $\boldsymbol{G}$ is right adjoint to $\boldsymbol{F}$, with ε as counit.

Dually, a selection of universal arrows to a functor $\boldsymbol{G}$ gives a left adjoint $\boldsymbol{F}$ to $\boldsymbol{G}$.

Proof. Lemma 6.2.5, Lemma 6.2.7 and Lemma 6.2.8 shows that a functor $\boldsymbol{F} : \mathcal{X} \longrightarrow \mathcal{A}$ and selected universal arrows from it give a unique right adjunct $\boldsymbol{G}$ for which the arrows form a natural transformation. $\square$

6.3 Comma Category

Definition 6.3.1 (Comma Category)**.** For categories $\mathcal{X}$ and $\mathcal{A}$ and a functor $\boldsymbol{F} : \mathcal{X} \longrightarrow \mathcal{A}$, the *comma category of* $\boldsymbol{F}$ *over* $\mathcal{A}$ (notation $(\boldsymbol{F} \downarrow \mathcal{A})$) is defined as a category where:

- Each object is a pair (X, f) with $\mathcal{X}$-object X and $\mathcal{A}$-morphism $f : \boldsymbol{F}X \longrightarrow A$:

$$X \qquad \boldsymbol{F}X \xrightarrow{\quad f \quad} A$$

- Each morphism from (X, f) to (X', f') is a pair (k, h) with $\mathcal{X}$-morphism $k : X \longrightarrow X'$ and $\mathcal{A}$-morphism $h : A \longrightarrow A'$ which makes the following square of $\mathcal{A}$ commute:

$$\begin{tikzcd}
X \arrow[d, "k"'] & \boldsymbol{F}X \arrow[r, "f"] \arrow[d, "\boldsymbol{F}k"'] & A \arrow[d, "h"] \\
X' & \boldsymbol{F}X' \arrow[r, "f'"'] & A'
\end{tikzcd}$$

Remark 6.3.2. Composition is defined by $(k, h) \circ (k', h') = (k \circ k', h \circ h')$, and identity arrows are (id_X, id_A).

Remark 6.3.3. We also use an alternative notation $(X, \boldsymbol{F}X \xrightarrow{f} A)$ for $(\boldsymbol{F} \downarrow \mathcal{A})$-object (X, f) to make explicit the domain and codomain of f.

Definition 6.3.4 (Comma Category (Dual))**.** Dually, for categories $\mathcal{X}$ and $\mathcal{A}$ and a functor $\boldsymbol{G} : \mathcal{A} \longrightarrow \mathcal{X}$, the *comma category of $\mathcal{X}$ over $\boldsymbol{G}$* (notation $(\mathcal{X} \downarrow \boldsymbol{G})$) is defined as a category where:

- Each object is a pair (g, A) with $\mathcal{X}$-morphism $g : X \longrightarrow \boldsymbol{G}A$ and $\mathcal{A}$-object A:

$$X \xrightarrow{\quad g \quad} \boldsymbol{G}A \qquad\qquad A$$

- Each morphisms from (g, A) to (g', A') is a pair (k, h) with $\mathcal{X}$-morphism $k : X \longrightarrow X'$ and $\mathcal{A}$-morphism $h : A \longrightarrow A'$ which makes the following square of $\mathcal{X}$ commute:

$$\begin{tikzcd}
X \arrow[r, "g"] \arrow[d, "k"'] & \boldsymbol{G}A \arrow[d, "\boldsymbol{G}h"] & A \arrow[d, "h"] \\
X' \arrow[r, "g'"'] & \boldsymbol{G}A' & A'
\end{tikzcd}$$

Remark 6.3.5. Composition and identity arrows are defined in the same way.

Definition 6.3.6 (Base functors)**.**

- The *base functor* $\boldsymbol{b} : (\mathcal{X} \downarrow \boldsymbol{G}) \longrightarrow \mathcal{X} \times \mathcal{A}$ takes an object $(f : X \longrightarrow \boldsymbol{G}A, A)$ to (X, A), and an arrow (k, h) to (k, h).
- The base functor $\boldsymbol{b}' : (\mathcal{F} \downarrow \mathcal{A}) \longrightarrow \mathcal{X} \times \mathcal{A}$ takes an object $(X, f : \boldsymbol{F}X \longrightarrow A)$ to (X, A), and an arrow (h, k) to (h, k).

Exercise 6.3.7. Verify that $\boldsymbol{b}$ and $\boldsymbol{b}'$ are indeed functors.

6.4 Adjunction as an isomorphism between comma categories

Theorem 6.4.1. Given an adjunction $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle$, there exists a unique functor isomorphism $\boldsymbol{I}$ that sends each $(X, f : \boldsymbol{F}X \longrightarrow A)$ to (g, A) where g is the adjunct of f and makes the following triangle commute:

$$\begin{tikzcd}
(\boldsymbol{F} \downarrow A) \arrow[rr, "\boldsymbol{I}"] \arrow[dr, "\boldsymbol{b}"'] & & (X \downarrow \boldsymbol{G}) \arrow[dl, "\boldsymbol{b}'"] \\
& X \times A &
\end{tikzcd}$$

Proof. We show that the left square commutes if the right square does, namely, if $k \circ f = f' \circ \boldsymbol{F}h$, then $\boldsymbol{G}k \circ g = g' \circ h$, and vise versa.

$$\begin{tikzcd}
X \arrow[d, "h"] & \boldsymbol{F}X \arrow[r, "f"] \arrow[d, "\boldsymbol{F}h"'] & A \arrow[d, "k"] & X \arrow[r, "g"] \arrow[d, "h"'] & \boldsymbol{G}A \arrow[d, "\boldsymbol{G}k"] & A \arrow[d, "k"] \\
X' & \boldsymbol{F}X' \arrow[r, "f'"'] & A' & X' \arrow[r, "g'"'] & \boldsymbol{G}A' & A'
\end{tikzcd}$$

This is equivalent to the fact that the adjunct of $k \circ f$ is $\boldsymbol{G}k \circ g$, and the adjunct of $f' \circ \boldsymbol{F}h$ is $g' \circ h$, which is proven as follows:

$$\begin{array}{rlcrl}
& \boldsymbol{G}(k \circ f) \circ \eta & \quad & & \epsilon \circ \boldsymbol{F}(g' \circ h) \\
= & \boldsymbol{G}k \circ \boldsymbol{G}f \circ \eta & & = & \epsilon \circ \boldsymbol{F}g' \circ \boldsymbol{F}h \\
= & \boldsymbol{G}k \circ g & & = & f' \circ \boldsymbol{F}h
\end{array}$$ $\square$

Exercise 6.4.2. Complete the above proof.

There is a converse to Theorem 6.4.1.

Theorem 6.4.3. Given a functor isomorphism $\boldsymbol{I} : (\boldsymbol{F}, A) \xrightarrow{\sim} (X, \boldsymbol{G})$ that commutes with a base category (X, A), $\langle \boldsymbol{F}, \boldsymbol{G}, \eta, \varepsilon \rangle$ is an adjunction where $\eta_X = \boldsymbol{I}(X, id_{\boldsymbol{F}X})$ and $\varepsilon_A = \boldsymbol{I}^{-1}(id_{\boldsymbol{G}A}, A)$. $\boldsymbol{I}$ and $\boldsymbol{I}^{-1}$ send arrows to their adjuncts.

Proof. It suffices to show that, for every $\mathcal{X}$-object X, $\boldsymbol{I}(X, id_{\boldsymbol{F}X})$ is universal from X to $\boldsymbol{G}$.

$(X, \boldsymbol{F}X \xrightarrow{id_X} \boldsymbol{F}X)$ is a $(\boldsymbol{F}, A)$-object and is sent to the $(X, \boldsymbol{G})$-object $(X \longrightarrow \boldsymbol{GF}X, \boldsymbol{F}X)$.

Consider any A-morphism $g : X \longrightarrow \boldsymbol{G}A$. Then $(X \xrightarrow{g} \boldsymbol{G}A, A)$ is a $(X, \boldsymbol{G})$-object. Let $(X, \boldsymbol{F}X \xrightarrow{g'} A)$ be $\boldsymbol{I}^{-1}(X \xrightarrow{g} \boldsymbol{G}A, A)$. Now consider the following diagram in $(\boldsymbol{F}, A)$:

$$\begin{tikzcd} X \arrow[d, "id_X"'] & \boldsymbol{F}X \arrow[r, "id_{\boldsymbol{F}X}"] \arrow[d, "\boldsymbol{F}id_X"'] & \boldsymbol{F}X \arrow[d, dotted, "g'"] \\ X & \boldsymbol{F}X \arrow[r, "g'"'] & A \end{tikzcd}$$

Thus (id_X, g') is an arrow in $(\boldsymbol{F}, A)$. By sending this diagram via $\boldsymbol{I}$, we obtain the following diagram in $(X, \boldsymbol{G})$ (note that $(X, \boldsymbol{F}X \xrightarrow{g'} A)$ is sent back to $(X \xrightarrow{g} \boldsymbol{G}A, A)$ by $\boldsymbol{I}$ since $\boldsymbol{I}$ is iso):

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[d, "id_X"'] & \boldsymbol{GF}X \arrow[d, "\boldsymbol{G}g'"] & \boldsymbol{F}X \arrow[d, "g'"] \\ X \arrow[r, "g"'] & \boldsymbol{G}A & A \end{tikzcd}$$

This is equivalent to the following diagram, which shows that $(\eta_X, \boldsymbol{F}X)$ is a universal arrow from X to $\boldsymbol{G}$ and g' is the adjunct of g.

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "g"'] & \boldsymbol{GF}X \arrow[d, "\boldsymbol{G}g'"] & \boldsymbol{F}X \arrow[d, "g'"] \\ & \boldsymbol{G}A & A \end{tikzcd}$$

The proof of uniqueness is left as an exercise. □

> **Theorem 6.4.4.** A universal arrow from $\boldsymbol{F}$ to A is an initial object of a comma category $(\boldsymbol{F}, \mathcal{X})$. Dually, a universal arrow from A to $\boldsymbol{F}$ is a terminal object of a comma category $(\boldsymbol{F}, \mathcal{X})$.

Exercise 6.4.5. Prove Theorem 6.4.4.

6.5 Adjuncts Compose

The composition of two adjunctions yileds an adjunction.

Theorem 6.5.1. For any four functors $\boldsymbol{F} : \mathcal{A} \longrightarrow \mathcal{B}$, $\boldsymbol{G} : \mathcal{B} \longrightarrow \mathcal{A}$, $\boldsymbol{F'} : \mathcal{B} \longrightarrow \mathcal{C}$ and $\boldsymbol{G'} : \mathcal{C} \longrightarrow \mathcal{B}$, if $\boldsymbol{F} \dashv \boldsymbol{G}$ and $\boldsymbol{F'} \dashv \boldsymbol{G'}$, then $\boldsymbol{F'F} \dashv \boldsymbol{GG'}$.

Proof. Consider the $\mathcal{A}$-arrow $\boldsymbol{G}\eta'_{\boldsymbol{F}X} \circ \eta_X$ in the following diagram:

$$\begin{tikzcd}
X \arrow[rr, "\eta_X"] \arrow[dr, dotted, "\boldsymbol{F}"'] & & \boldsymbol{GF}X \arrow[rrr, "\boldsymbol{G}\eta'_{\boldsymbol{F}X}"] & & & \boldsymbol{GG'F'F}X \\
& \boldsymbol{F}X \arrow[ur, dotted, "\boldsymbol{G}"] \arrow[rrr, "\eta'_{\boldsymbol{F}X}"] \arrow[dr, dotted, "\boldsymbol{F'}"'] & & & \boldsymbol{G'F'F}X \arrow[ur, dotted, "\boldsymbol{G}"] & \\
& & \boldsymbol{F'F}X \arrow[urr, dotted, "\boldsymbol{G'}"'] & & &
\end{tikzcd}$$

For any $\mathcal{A}$-arrow $f : X \longrightarrow \boldsymbol{GG'}A$, its adjunct $g : \boldsymbol{F}X \longrightarrow \boldsymbol{G'}A$ with respect to $\boldsymbol{F} \dashv \boldsymbol{G}$ is a $\mathcal{B}$-arrow such that $f = \boldsymbol{G}g \circ \eta_X$, and g's adjunct $h : \boldsymbol{F'F}X \longrightarrow A$ with respect to $\boldsymbol{F'} \dashv \boldsymbol{G'}$ is a $\mathcal{C}$-arrow such that $g = \boldsymbol{G'}h \circ \eta'_{\boldsymbol{F}X}$. Thus,

$$\begin{aligned}
f &= \boldsymbol{G}g \circ \eta_X \\
&= \boldsymbol{G}(\boldsymbol{G'}h \circ \eta'_{\boldsymbol{F}X}) \circ \eta_X \\
&= \boldsymbol{GG'}h \circ \boldsymbol{G}\eta'_{\boldsymbol{F}X} \circ \eta_X
\end{aligned}$$

Thus, $(\boldsymbol{G}\eta'_{\boldsymbol{F}X} \circ \eta_X, \boldsymbol{F'F}X)$ is an universal arrow from X to $\boldsymbol{GG'}$.

The rest of the proof consists of checking that $\boldsymbol{F'F}$ sends $\mathcal{A}$-morphism $f : A \longrightarrow B$ to the adjunct of $\boldsymbol{G}\eta'_{\boldsymbol{F}B} \circ \eta_B \circ f$. This is left as an exercise. $\square$

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